



Proposal Report

by

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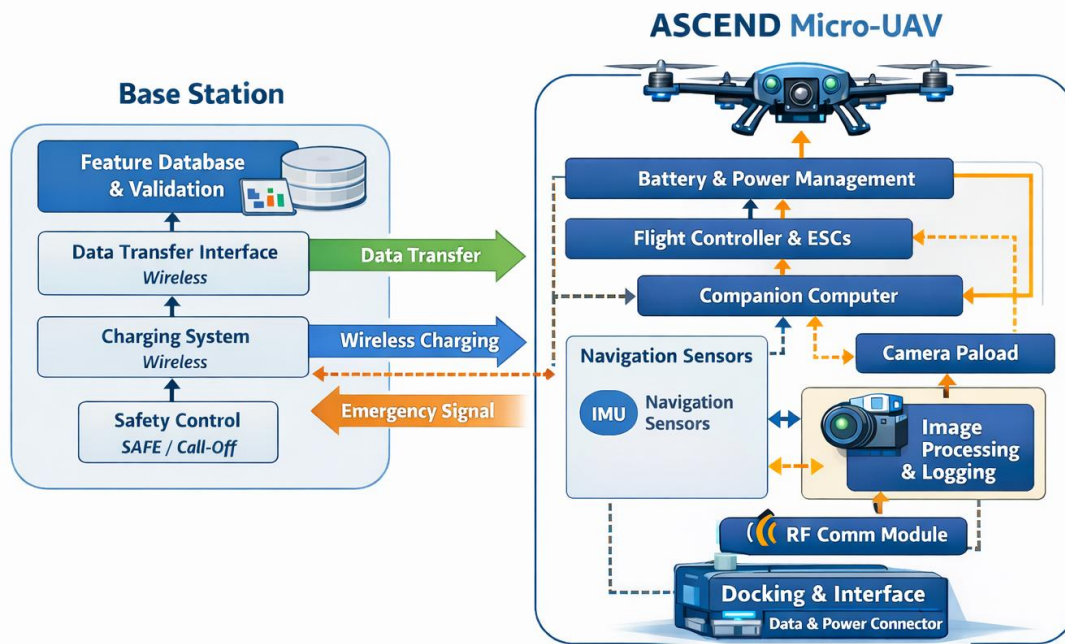
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1.Description of ASCEND

The **ASCEND (Towards Swarm Expedition: Autonomous Surveyor Challenge for Exploration, Navigation and Dynamics)** project aims to develop and demonstrate a **fully autonomous micro-UAV system** capable of navigation, survey, data transfer, and recharging in **GNSS-denied environments**, in alignment with the objectives of the **ISRO Robotics Challenge – URSC (IRoC-U 2026)**.

Future planetary exploration missions, including **Martian surface reconnaissance**, require aerial platforms that can operate independently of external navigation infrastructure while executing repeated autonomous sorties. Addressing this need, the proposed system consists of a **micro-UAV and a base station**, designed to function as a closed autonomous loop without human intervention.



ASCEND SYSTEM OVERVIEW – IRoC-U 2026

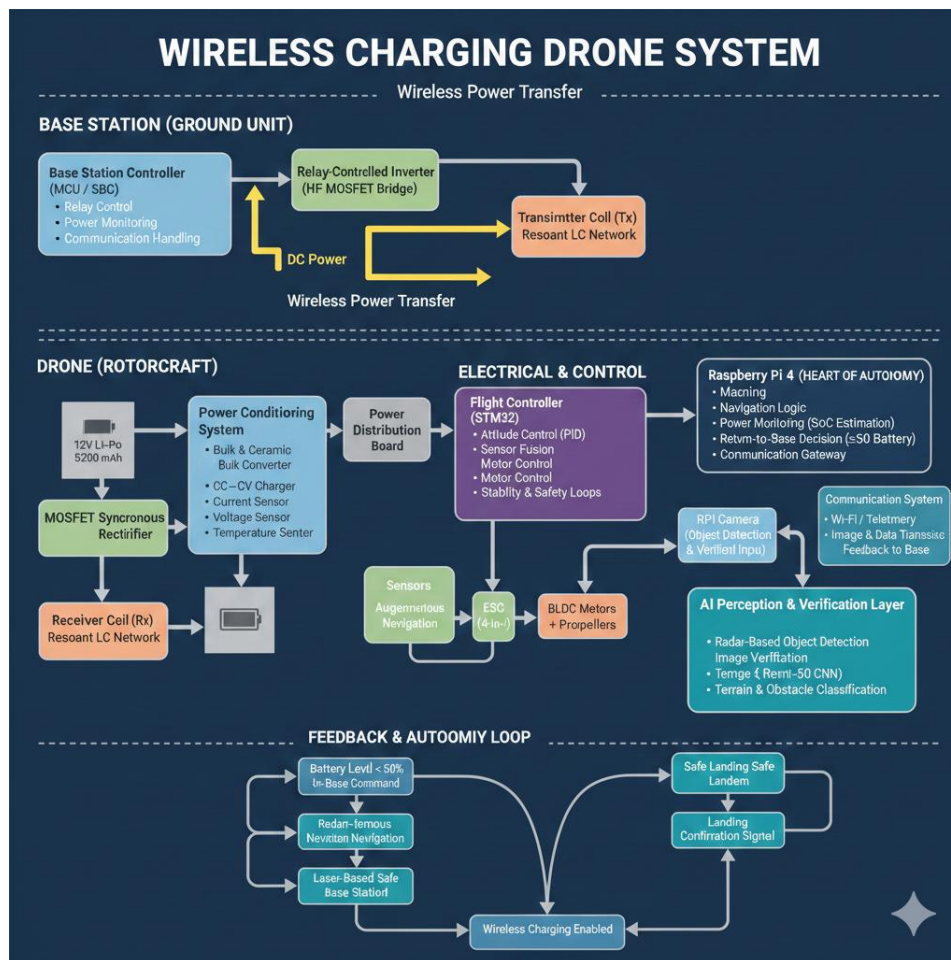
The UAV employs a **radar-assisted SLAM (Simultaneous Localization and Mapping)** framework for navigation and guidance. Radar sensing provides robust perception under varying environmental conditions, while SLAM enables local map generation and real-time pose estimation relative to the base station, allowing reliable autonomous navigation without GNSS, pseudolites, or reflector arrays.

During operation, the UAV autonomously departs from the base station, acquires reference visual features, surveys the designated arena to detect target features, records their relative locations, and captures verification images. Upon mission completion, the UAV autonomously returns to the base station and performs precision landing and docking, followed by **wireless data transfer and wireless power charging**.

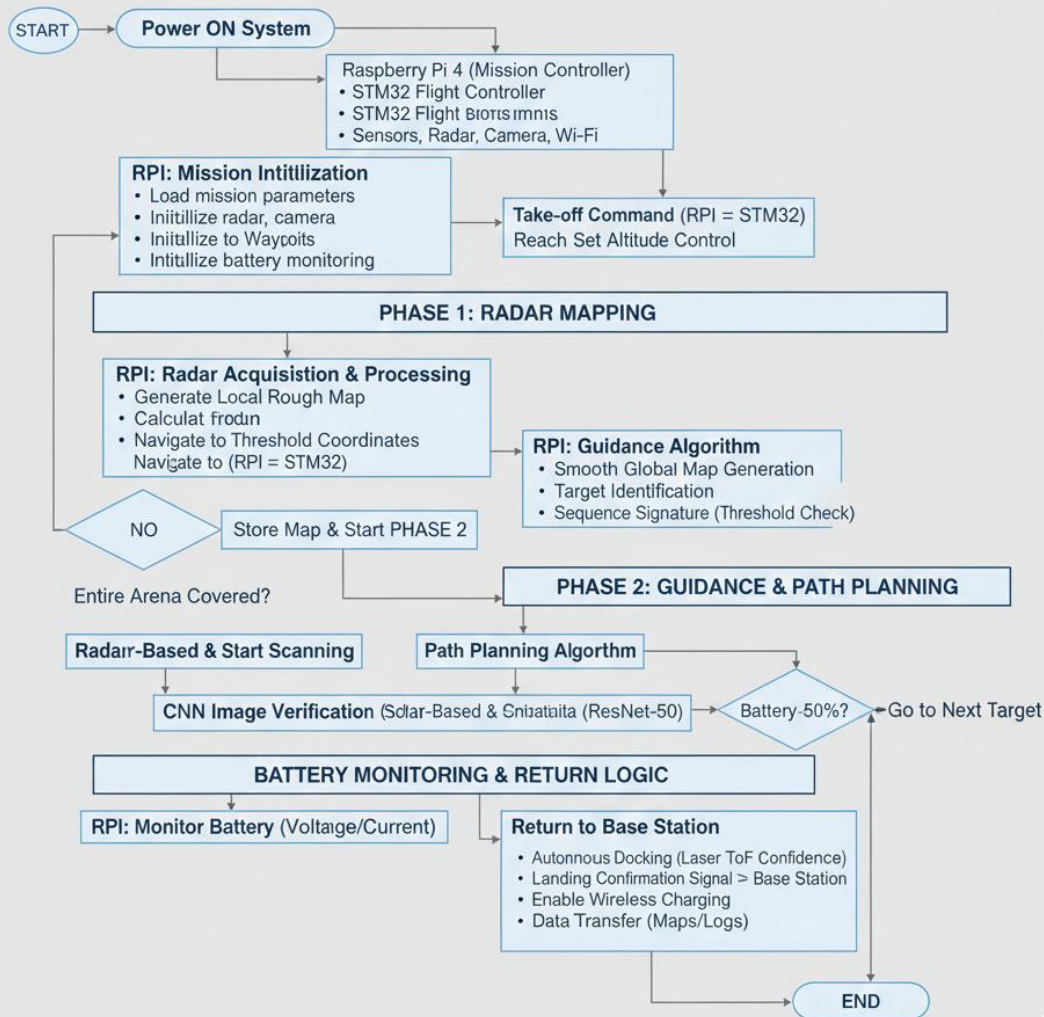
Wireless charging is implemented using **inductive coupling between two aligned coils**, with a transmitter coil integrated into the base station and a receiver coil mounted on the UAV. An alternating current in the transmitter coil generates a magnetic field that induces voltage in the receiver coil, which is rectified and regulated to safely charge the onboard battery.

The base station validates the received image data against a reference feature database, enabling multiple autonomous sorties. This project emphasizes **autonomy, navigation, and system integration**, demonstrating a scalable foundation for future **autonomous aerial swarm systems** for planetary exploration.

2. System Architecture



Wireless Charging Drone System Logic Flow



3. Identification of components with their specifications

a) Mechanical Subsystem: Structure and Propulsion

The mechanical structure of the rotorcraft is based on the F450 quadcopter frame, which follows a standard X-type configuration. This configuration provides uniform load distribution across all four arms and ensures balanced thrust generation during hover and descent. The frame consists of a central load-

bearing plate assembly and detachable arms, which simplifies mechanical assembly and allows easy replacement in case of damage during testing. Compared to a fully custom-machined or 3D-printed structure, the F450 frame offers better structural repeatability and sufficient stiffness to handle dynamic flight loads, vibrations, and landing impacts.

For propulsion, the quadcopter uses fixed-pitch propellers manufactured from ABS and nylon composite material. ABS contributes to low mass and ease of manufacturing, while nylon improves toughness, fatigue resistance, and impact tolerance. This combination allows the propellers to withstand repeated cyclic loading and minor ground contact without brittle failure. When compared to basic plastic propellers, the ABS–nylon propellers show improved durability, and when compared to carbon-fiber propellers, they offer lower cost and safer failure behavior, which is preferred for experimental and competition-based platforms.

Overall, the combination of the F450 structural frame and ABS–nylon fixed-pitch propellers provides a mechanically stable, lightweight, and reliable platform that supports precise control, ease of integration, and repeated operational testing required for the proposed landing system.

b) Battery and Recharging

Battery selection and charging strategy are designed to balance energy density, discharge capability, safety, and longevity, recognizing the inherent efficiency limitations of wireless power transfer. Lithium-polymer (Li-Po) batteries are chosen for their high power-to-weight ratio and ability to supply the burst currents required for drone propulsion, but they demand carefully controlled charging due to their sensitivity to voltage, current, and temperature extremes.

A primary trade-off in the system is between charging speed and efficiency. Wireless charging introduces coupling and switching losses, and aggressive fast charging significantly increases thermal stress on the coils, MOSFETs, and battery. To address this, the system adopts a moderate charging rate, prioritizing thermal safety and efficiency over rapid recharging. This approach is well-suited to planetary missions, where reliability and component lifespan are more critical than minimizing downtime.

The choice of wireless charging over wired charging further reflects a deliberate trade-off. Although wired charging is more efficient, exposed connectors are vulnerable to dust contamination and mechanical misalignment. Wireless charging sacrifices some efficiency in exchange for dust tolerance, electrical isolation, and mechanical robustness, making it more suitable for long-term Mars surface operations.

Thermal management also favors simplicity and reliability. Instead of active cooling, the system relies on passive thermal design and controlled power levels. This limits peak charging power but reduces system complexity, mass, and failure points, reinforcing the overall emphasis on safe and reliable operation.

Final Insight:-

The combined design of adaptive wireless power transfer, relay-controlled activation, and conservative battery charging represents a systems-level optimization. By trading charging speed and peak efficiency for safety, autonomy, and environmental resilience, the system achieves a practical and deployable solution for autonomous drone operations in harsh planetary environments.

c) Algorithms:-

- **Base Station Detection & World Frame Anchoring**
Algorithm: CFAR-based peak detection with SNR thresholding
Provides a reliable global reference with low computational cost but requires a strong radar-reflective base station and is sensitive to clutter.
 - **Sensor Fusion for Pose Estimation**
Algorithm: Extended Kalman Filter (EKF)
Enables accurate 6-DoF pose estimation under uncertainty at the cost of higher computation and careful noise tuning.
 - **Arena Exploration Strategy**
Algorithm: Boustrophedon Coverage Path Planning
Ensures uniform coverage of structured arenas with low overhead but lacks adaptability to irregular environments.
 - **Coordinate Transformation**
Algorithm: 3D Rigid Body Transformation
Provides exact sensor-to-world mapping but is sensitive to localization errors.
 - **3D Map Representation**
Algorithm: Voxel-Based Probabilistic Mapping (OctoMap)
Efficiently models sparse radar data and uncertainty but results in coarse spatial resolution.
 - **Pose Correction and Drift Reduction**
Algorithm: NDT Scan Matching
Robust to sparse and noisy radar scans but requires higher computational resources.
 - **Navigation and Obstacle Avoidance**
Algorithm: A* / RRT Path Planning
Generates reliable collision-free paths but may not be globally optimal in complex environments.
 - **Object Detection from Radar Data**
Algorithm: DBSCAN Clustering
Detects objects without prior count information and handles noise well, but requires careful parameter tuning.
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- A pre-trained ResNet-50 was chosen over lightweight CNNs (e.g., MobileNet, VGG-16) and traditional feature extractors.

Higher computational and memory cost versus superior representational capacity.

ResNet-50's residual connections mitigate vanishing gradients and allow effective training on deep feature hierarchies. This is critical when learning non-natural textures present in radar-projected images, where edge continuity and reflectivity patterns differ significantly from optical imagery.

d) Control System and Sensor Architecture

The flight performance of an autonomous UAV is fundamentally limited by the precision of its sensor suite and the responsiveness of its processing unit. For this experimental build, the hardware stack was selected to support high-fidelity **SLAM (Simultaneous Localization and Mapping)** and custom flight dynamics.

>>Flight Controller Rationale

While commercial off-the-shelf (COTS) flight controllers offer stability, they lack the modularity required for experimental autonomous navigation.

- **Custom Firmware Implementation:** We have opted to develop a custom flight controller using the **PX4 Autopilot** framework. This allows for the removal of unnecessary modules (like GPS-dependency) and enables direct manipulation of stability derivatives.
- **MCU Selection:** The **STM32F407VGT6** was selected over the higher-spec F7 series. It provides the optimal balance of computational overhead for sensor fusion while maintaining lower power consumption and thermal profiles, crucial for extending flight endurance.

>>Sensor Tradeoff Analysis

To support radar-based navigation with a 3.5 cm range resolution, the sensor stack was upgraded from "standard" drone components to high-precision alternatives.

A. Inertial Measurement Unit (IMU)

We transitioned from the industry-standard MPU-6050 to the **ICM-42688-P**.

- **Tradeoff:** Higher unit cost vs. superior vibration immunity.
- **Justification:** The ICM-42688-P offers significantly lower noise floors and higher data rates, which are critical for stable flight mechanics during precise SLAM mapping.

B. Barometric Altimeter

Precision altitude hold is required to complement radar data. The **Bosch BMP390** was selected over the legacy BMP280.

- **Key Advantage:** A 75% reduction in relative altitude error (0.25m vs 1.0m).
- **Vertical Stability:** The ultra-low RMS noise (0.02 Pa) eliminates "vertical jitter" during hover, providing a stable baseline for the complementary filter to fuse with radar readings.

C. Magnetometer and Interference Mitigation

The **Bosch BMM150** was selected as the primary heading sensor over the QMC5883L due to the specific requirements of the wireless charging system.

- **Shielding Strategy:**
 - **AC Field Mitigation:** Ferrite sheeting on charging coils channels high-frequency flux.
 - **DC Field Protection:** Mu-Metal side-walls protect the sensor from local battery magnetism while maintaining exposure to the Earth's magnetic field.
 - **Spatial Isolation:** Sensors are placed in a central "quiet zone," maximizing distance ($1/r^3$ law) from perimeter-mounted charging coils.
- **Tradeoff: BMM150 vs. QMC5883L:**
 - **Resolution:** The BMM150 offers 0.3 μ T resolution. In a shielded "quiet zone" where signals are attenuated, this higher sensitivity is critical to distinguish the Earth's magnetic field from the noise floor.
 - **Saturation Handling:** The BMM150 features a wide dynamic range 2500 μ T on the Z-axis), preventing sensor "pegging" during momentary charging surges.
 - **Thermal Stability:** Given that wireless charging generates heat, the BMM150's superior temperature stability prevents heading drift as components warm up.

>> Hardware Selection Matrix

Component	Selected Model	Primary Justification
MCU	STM32F407	Optimized power/thermal footprint for long-duration flight.
IMU	ICM-42688-P	High-speed, low-noise 6-DOF essential for SLAM.
Barometer	BMP390	24-bit resolution for precise altitude hold (± 0.25 m).
Magnetometer	BMM150	High resolution and saturation resistance in shielded environments.

4.ASCEND Realization plan

SI No	Hardware details	Procurement source (Market /Fabrication/3D printing)	Specification/Realisation plan	Quantity
1.	BLDC motor	market	Rotation of the propeller	4
2.	Frame	market	3D printed 450mm polyamide nylon/to mount all the components	1
3.	ESC(4 in 1)	market	Controlling the speed of motor	1
4.	Lipo Battery	market	5000mAh-3S-12V for power supply	2
5.	Raspberry Pi 4	market	Main brain of the system for communication and control and as companion computer, AI, vision, navigation	2
6.	Raspberry Pi camera	market	For capturing and processing image	1
6.	Microcontrollers STM 32	market	Heart of the system to control all the hardware parts	1
7.	IMU-ICM-42688P	market	6 axis degree of freedom gyroscope and accelerometer	1
8.	Barometer-BMP390	market	Converts pressure into altitude	1
9.	Magnetometer-BMM150	market	To control the yaw motion of the UAV	1
10.	Laser module(Time of flight sensor)	market	For precise landing of UAV on base station	4
11.	Buck converter	market	To step down the dc voltages according to the need of the sensors and microcontroller	3
12.	Coils	market	For primary and secondary coil of wireless charging	As required
13.	MOSFETs	market	High frequency switching in wireless charging	4
14.	Relay module	market	For efficient use of power	1
15.	Capacitors	market	For resonance and the stability of voltage Specification 1000 μ F – 2200 μ F	1
16.	Radar IWR6843A0PEVM	market	For navigation and mapping	1

5. Testing Plan

The testing strategy is designed around:

- Incremental risk reduction
- Early fault detection
- Safe integration of hardware and software

Testing is divided into subsystem validation, simulation-based testing, controlled flight testing, mission execution, and safety evaluation. Advancement between phases is based on predefined acceptance criteria.

Phase 1: Subsystem-Level Testing

1 Sensor Subsystem Testing

Onboard sensors (radar, IMU, magnetometer, and barometer) are tested individually to verify data integrity, accuracy, and stability.

- Radar performance is evaluated for range consistency, detection reliability, and update rate.
- IMU and magnetometer outputs are monitored for bias, noise, and drift.
- Barometer readings are tested for repeatability under static conditions.

Acceptance Criteria: Stable data output within acceptable error limits.

2 Onboard Computing Testing (Raspberry Pi)

The Raspberry Pi onboard computer is tested for boot reliability, computational stability, and communication with peripheral devices.

- CPU and memory usage are monitored under full autonomy stack load.
- Inter-process communication between navigation, perception, and control modules is verified.

Acceptance Criteria: Continuous stable operation without crashes or delays.

3 Flight Controller Testing (STM32)

The flight controller is tested independently for attitude stabilization, command response, and failsafe execution.

- Control signal response is verified.
- Emergency modes and motor cut-off logic are validated.

Acceptance Criteria: Deterministic and safe response to control inputs.

4 Wireless Charging System Testing

Wireless charging is tested under varying coil alignments.

- Power transfer efficiency and thermal behavior are measured.
- Battery charging safety and regulation are verified.

Acceptance Criteria: Reliable charging without overheating or instability.

Phase 2: Software-in-the-Loop (SITL) Testing

Software-in-the-Loop testing is conducted to validate autonomy algorithms without flight risk.

- UAV dynamics are simulated.
- Radar-SLAM navigation, waypoint generation, and return-to-base logic are tested.
- Failure cases such as localization loss and low battery are injected.

Acceptance Criteria: Successful completion of simulated missions.

Phase 3: Hardware-in-the-Loop (HITL) Testing

In HITL testing, real hardware operates within a simulated environment.

- Raspberry Pi, flight controller, and sensors are integrated.
- Timing synchronization and real-time performance are evaluated.

Acceptance Criteria: Correct actuator commands with stable timing behavior.

Phase 4: Controlled Indoor Flight Testing

Indoor flight testing is conducted in a GNSS-denied environment.

1 Basic Flight Tests

- Autonomous takeoff and hover
- Altitude and attitude stability

2 Navigation Tests

- Short-range waypoint following
- Radar-SLAM localization consistency
- Drift estimation and correction

Acceptance Criteria: Stable flight with acceptable localization error.

Phase 5: Vision and Data Acquisition Testing

The onboard camera system is tested for image quality, synchronization, and storage reliability.

- Image capture under different lighting conditions
- Timestamp and metadata logging verification

Acceptance Criteria: Clear images suitable for feature validation with no data loss.

Phase 6: Autonomous Docking and Wireless Charging

Precision landing and docking with the base station are tested.

- Autonomous approach and landing alignment
- Automatic initiation of wireless charging
- Charging status feedback verification

Acceptance Criteria: Successful docking and charging without human intervention.

Phase 7: Mission-Level Task Testing

Competition-specific tasks are validated:

- Seeding of reference images
- Autonomous search and survey
- Feature detection and coordinate logging
- Return, docking, data transfer, and recharge

Acceptance Criteria: Successful completion of full mission sequence.

Phase 8: End-to-End Autonomous Sorties

Multiple autonomous sorties are conducted to evaluate system repeatability.

- Performance consistency across sorties
- Battery endurance and recharge reliability

Acceptance Criteria: Repeatable mission success without manual intervention.

Phase 9: Safety and Failure Mode Testing

Emergency scenarios are intentionally triggered:

- Communication loss
- Sensor degradation
- Low battery conditions
- Compute failure

The system response is evaluated for safety and predictability.

Acceptance Criteria: Controlled recovery or safe landing in all cases.

Performance Metrics

- Localization accuracy (relative frame)
 - Mission success rate
 - Docking accuracy
 - Wireless charging efficiency
 - Data transfer reliability
 - System repeatability
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Conclusion

This testing plan ensures the ASCEND UAV system is safe, reliable, and mission-ready. The phased approach validates autonomy, navigation, docking, and charging capabilities while minimizing risk. Successful execution demonstrates readiness for IRoC-U 2026 and future autonomous planetary exploration missions.

6. Brief of system specification

Sl. No.	Description	Specification
1.	Overall mass	1.8 Kg
2.	Overall dimensions	450mm, 450mm, 200mm (approx.)
3.	Power requirements	60-70 Watt, 12 Volt, 5.2 Ampere (approx.)
4.	Flight time per charge	15-20 minutes
5.	Number of propellers and their size	4 propellers, 10.45 inch diameter
6.	Autonomy Features	Autonomous takeoff, navigation, survey, target detection, obstacle avoidance, return, docking, charging
7.	Navigation Method	Radar-assisted SLAM-based navigation
8.	Onboard Computing	Raspberry Pi 4
9.	Flight Control Unit	STM-32 based development board
10.	Vision System	Raspberry Pi camera module for image acquisition and object detection
11.	Communication	Wireless communication between UAV and base station, UART onboard communication
12.	Sensor Suite	Radar sensor, IMU, Magnetometer, Barometer
13.	Autonomous Landing	Laser and vision assisted autonomous landing on the base station
14.	Charging Method	Inductive wireless power transfer using aligned coils
15.	Safety Features	Autonomous landing, return to base for controlled mission abort
16.	Data Handling	Onboard image capture, logging, wireless data transfer to base station

7. Overview of the emergency response system.

Autonomous UAV operation in **GNSS-denied environments** requires a reliable emergency handling mechanism to ensure safe behaviour under abnormal conditions. The ASCEND UAV performs autonomous navigation, survey, docking, wireless charging, and data transfer without human intervention, making an integrated fail-safe system essential.

Identification of Emergency Situations

The following emergency scenarios are anticipated during task execution:

- **Navigation failure:** Loss of localization or excessive drift in radar-based SLAM.
- **Sensor degradation:** Radar or inertial sensor malfunction affecting state estimation.
- **Communication loss:** Interruption of wireless link with the base station.
- **Power anomaly:** Low battery during flight or failure to initiate wireless charging.
- **Onboard compute failure:** Crash or hang of the Raspberry Pi autonomy stack.
- **Unsafe flight condition:** Excessive attitude deviation or unstable flight.

Proposed Emergency Response System

ASCEND employs a **hierarchical emergency response architecture** with high-level decision making on the **Raspberry Pi** and low-level safety enforcement on the **STM32 flight controller**.

The system continuously monitors sensor health, localization confidence, power status, communication link, and flight stability. Upon detecting an anomaly, failures are classified as recoverable or critical. Appropriate responses are automatically triggered, including:

- **Return-to-Base (RTB)** when navigation is reliable but mission continuation is unsafe.
- **Autonomous landing** during low-battery or localization failure.
- **Hover and stabilize** for temporary sensor or communication faults.
- **SAFE mode / emergency landing** executed by the flight controller in critical failures.

This layered approach ensures safe recovery even if high-level autonomy fails.

Safety Features and Redundancy

- Multiple fail-safe modes: RTB, autonomous landing, hover, SAFE mode
- Separation of autonomy (Raspberry Pi) and flight stabilization (STM32)
- Independent power monitoring and control enforcement by the flight controller

Alternative Systems and Rationale

Manual remote intervention and external navigation aids were not considered due to competition constraints. Full hardware redundancy was avoided to minimize mass, power consumption, and system complexity. The selected approach provides an optimal balance between **safety, simplicity, and robustness**, making it suitable for autonomous planetary exploration scenarios.

8. Project management

No.	Task	Main Responsibility	Deadline for Completion	Secondary Responsibility
1	Wireless charging and base station	Sudipta Pal	15.03.2026	Sachin Kumar
2	Navigation And image processing	Rohan	18.03.2026	Sonu
3	Assembly Testing and Validation	Hemanth	25.03.2026	Pranav
4	Flight Dynamics and stability	Pranav	25.03.2026	Sonu

9. Novelty in the overall proposal

Novelty of the Proposed System:-

This work proposes a radar-only UAV 3D mapping and target localization system that constructs real-time spatial maps using purely numerical FMCW radar data, eliminating cameras and LiDAR. Raw radar samples are processed into a range–Doppler–angle cube, elevation is estimated using inter-antenna phase differences, and detections are fused into a voxel-based occupancy map in the UAV frame.

Radar-only 3D mapping has remained largely unexplored because radar data is non-human-readable, height estimation is highly phase-sensitive, multipath ghosting dominates indoor scenes, and no standard radar-SLAM pipelines exist.

Compared to conventional vision-based methods, the proposed approach is invariant to lighting, fog, smoke, rain, and dust, does not capture identifiable imagery, enables stealth operation, and maintains reliable navigation in environments where camera and LiDAR systems fundamentally fail.

Novelty in the landing system at the base station:-

Precision Landing Using Vision and Multi-Point ToF Sensors

In this proposal, we introduce a precision landing mechanism for the quadcopter which mainly focuses on achieving proper centering and tilt-free landing on the base station. The idea is based on a combination of computer vision and Time-of-Flight (ToF) sensors, supported by a raised mechanical landing structure in the shape of a plus sign.



Normally, drones land using GPS, vision only, or a single height sensor, which can lead to tilt or misalignment during landing. To avoid this, we have designed a system where the drone decides when to descend based on multiple distance readings, instead of just altitude.

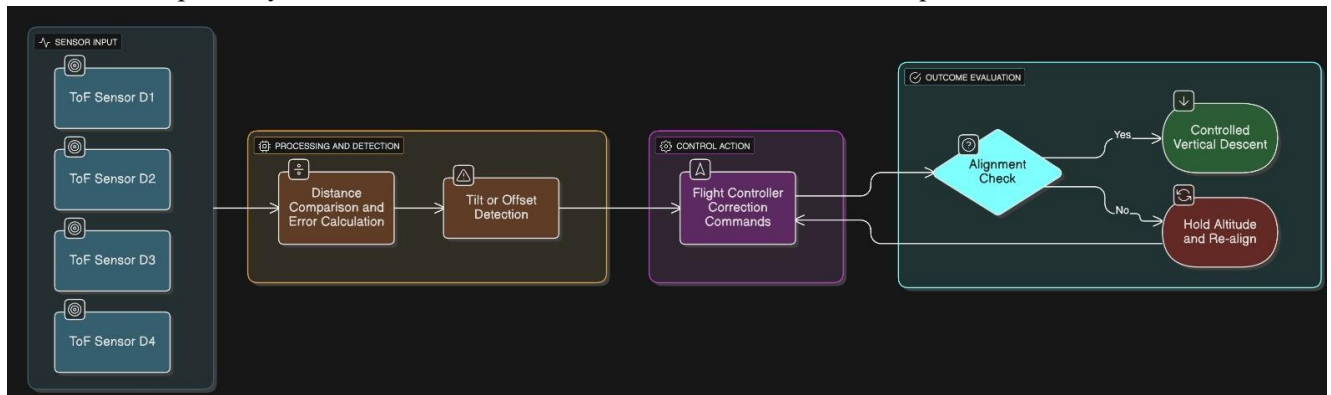
Working Concept

Initially, when the drone is at a height of around 3 to 5 meters, it uses a downward-facing camera to detect the plus-shaped marker placed on the base station. Using basic computer vision techniques, the drone tries to align itself near the center of this marker. At this stage, the alignment does not need to be perfect and is only meant for coarse positioning.

After reaching a lower altitude, the vision system is no longer reliable due to lighting variations and shadows. Hence, the control is shifted to four VL53L0X ToF sensors, mounted underneath the drone, one near each landing gear, all facing downward.

A raised plus-shaped structure is provided on the base station so that the ToF sensors get consistent and reliable distance measurements. The drone continuously checks the distance values from all four sensors.

If the drone is perfectly centered and level, all four distances will be almost equal.



If the drone tilts or shifts sideways, the distance values become unequal.

Based on this difference, the drone actively opposes the tilt by applying corrective roll and pitch commands. The drone is allowed to descend only when all four distance values are within a small

tolerance range. Otherwise, it holds its position, corrects itself, and then continues descending. This logic helps in achieving a stable and precise landing.

Why This Is Novel

- Landing decision based on distance equality, not only altitude
- Use of four ToF sensors instead of one, improving tilt detection
- Raised plus-shaped landing structure improves sensor reliability
- Vision used only where it is effective, ToF used for final precision
- Simple logic, but very effective for accurate landing
- Power Transfer – Novelty

Wireless Charging System:-

The proposed wireless charging system extends resonant inductive power transfer from conventional static applications to autonomous aerial drones operating in unstructured planetary environments. Instead of exposed electrical contacts, power is transferred through a magnetic field, eliminating mechanical connectors that are highly susceptible to dust, corrosion, and wear. This connector-less approach is particularly suited to Martian conditions, where fine dust, extreme temperature variations, and limited maintenance make wired charging unreliable.

A key novelty of the system is its use of adaptive resonant power transfer rather than fixed-frequency operation. In real deployments, perfect alignment between the transmitter coil on the base station and the receiver coil on the drone cannot be guaranteed due to variations in landing angle, height, and lateral offset. The system dynamically adjusts switching frequency, duty cycle, and inverter drive strength to maintain resonance under these conditions, ensuring stable power transfer without requiring precise mechanical positioning.

Charging is further controlled through an intelligent power regulation mechanism. The base station monitors reflected impedance and received power to estimate coupling efficiency and detect abnormal conditions such as misalignment or foreign objects. Power transmission is reduced or interrupted when inefficiencies are detected, preventing overheating and improving overall safety. Additionally, a relay-controlled coil activation mechanism is implemented: the transmitter coil is energized only after the drone has landed and explicitly sends a confirmation signal to the base-station controller. This ensures that wireless power is delivered only during stable ground contact, eliminating unnecessary power loss and preventing unsafe mid-air coupling.

The system is also optimized for Mars-like environments with minimal convective cooling. Lower resonant operating frequencies are used to reduce conductor losses, while ferrite shielding and controlled magnetic flux paths minimize stray fields and thermal stress. Together, these design choices enable reliable, long-duration operation in harsh planetary conditions.

Overall, the novelty lies not merely in wireless charging, but in the realization of a robust, adaptive, and autonomous power transfer system that safely delivers energy to aerial robots without human intervention or mechanical connectors.

